

CIS 6261: Trustworthy Machine Learning

Final Project Report [Option 1]: Practical Defenses Against Membership Inference and Adversarial Attacks Without Retraining

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Abstract

Machine learning models face dual threats from adversarial attacks and privacy leakage. This project develops practical inference time defenses that protect pre-trained models without retraining. We combine input preprocessing (feature squeezing) with output sanitization (temperature scaling, confidence clipping, noise injection) to simultaneously improve adversarial robustness and privacy. On CIFAR-10, our defense achieves 786% improvement in adversarial accuracy and 94.6% reduction in membership inference advantage, with 16% clean accuracy tradeoff. On CIFAR-100 with adversarial training, we achieve 139.6% adversarial improvement with only 8.7% accuracy drop, demonstrating that training time and inference time defenses are complementary.

1 Introduction

Machine learning models that are deployed in the real world applications face significant security and privacy vulnerabilities. Adversarial attacks craft imperceptible input perturbations to induce misclassification [1], while membership inference attacks exploit model outputs to determine whether specific data points were included in the training set [2]. These vulnerabilities present serious risks in domains such as healthcare, finance, and biometric authentication, where both prediction accuracy and data protection are critical.

Traditional defenses such as adversarial training [3] and differential privacy [4] provide robust protection but require computationally expensive retraining from scratch. This approach is often quite infeasible for production systems where models are already deployed, training data may no longer be accessible, or computational resources are limited. This project addresses the practical question: *Can we protect pre-trained models using only inference time modifications?*

We investigate two complementary settings. First, we defend a fixed CIFAR-10 classifier using only inference time defenses, without any model modification or retraining. Second, we train a CIFAR-100 model with adversarial training incorporated during training, then apply additional inference time defenses. This two-part approach allows us to evaluate both the standalone effectiveness of inference time defenses, and their synergy with training time robustness techniques.

Our defense strategy combines input preprocessing to mitigate adversarial perturbations with output sanitization to reduce privacy leakage. We demonstrate that this layered approach achieves substantial improvements in both adversarial robustness (up to 786% improvement) and privacy protection (up to 94.6% reduction in membership inference advantage), though with necessary accuracy tradeoffs. Our results provide practical guidance for organizations balancing security, privacy, and utility in deployed machine learning systems.

2 Background & Related Work

2.1 Adversarial Attacks and Defenses

Adversarial examples are inputs that are crafted by adding imperceptible perturbations that cause misclassification [1]. Modern gradient based attacks exploit backpropagation to efficiently compute perturbations. FGSM [5]

performs a single gradient step in the direction that maximizes loss. PGD [3] iteratively refines perturbations with projection onto an ℓ_∞ ball, making it significantly stronger. C&W [6] formulates attack generation as a constrained optimization problem to find minimal ℓ_2 perturbations.

Defense strategies fall into several categories. **Adversarial training** [3] augments training data with adversarial examples, creating inherently robust models but requiring complete retraining with 5–10 $\times$ computational overhead. **Input transformation defenses** such as feature squeezing [7] reduce input complexity through bit depth reduction and spatial filtering to remove adversarial perturbations without model modification. **Randomized smoothing** [8] provides provable robustness guarantees by averaging predictions over noisy inputs, trading accuracy for certified robustness radii.

However, Athalye et al. [9] demonstrated that many defenses relying on obfuscated gradients provide false security against adaptive attacks that account for the defense mechanism. This motivates our focus on defenses with clear operational principles rather than gradient masking.

2.2 Membership Inference Attacks and Defenses

Membership inference attacks [2] exploit the tendency of models to memorize training data, exhibiting higher confidence on training samples compared to test samples. These attacks threaten privacy by revealing whether specific individuals’ data was used for training, potentially violating regulations such as GDPR or HIPAA.

The basic attack strategy uses a threshold on the confidence scores or prediction entropy to classify samples as members or non-members. More sophisticated attacks train shadow models to learn the membership signal or use features beyond confidence, such as loss values or prediction stability [10].

Defenses include **differential privacy** [11], which adds calibrated noise during training to bound worst case privacy leakage but requires retraining and degrades utility. **Prediction sanitization** [12] post processes outputs to reduce information leakage without model modification. Temperature scaling reduces overconfidence by dividing logits by a temperature parameter before softmax. Confidence clipping caps maximum prediction probabilities to limit membership signals.

2.3 Inference Time Defense Strategies

Our work focuses on inference time defenses applicable to pre-trained models without modification. We adapt and combine techniques from feature squeezing [7], temperature scaling, confidence clipping [12], and output perturbation to address both adversarial robustness and privacy simultaneously. This setting is practically motivated but technically constrained: training time techniques such as adversarial training or differentially private SGD are unavailable for Part 1, though we incorporate them in Part 2 to demonstrate complementarity.

The main challenge is balancing the three competing objectives: maintaining utility (accuracy on clean inputs), achieving robustness (accuracy on adversarial inputs), and preserving privacy (low membership inference advantage). Our layered defense addresses each objective through targeted interventions at different stages of the inference pipeline.

3 Approach

3.1 Part 1: Defending a Pre-trained CIFAR-10 Model

3.1.1 Problem Setup and Baseline Vulnerabilities

We were provided with a pre-trained convolutional neural network trained on CIFAR-10 (32 $\times$ 32 RGB images, 10 classes). The baseline model achieved 93.44% training accuracy and 85.74% validation accuracy, which indicates slight overfitting. The model exhibited severe vulnerabilities:

- **Adversarial vulnerability:** Only 6.80% accuracy on provided adversarial examples (advexp0.npz), indicating near complete failure under attack
- **Privacy leakage:** Membership inference advantage of 0.261, substantially above random guessing (advantage 0)

Our objective was to design an inference time defense that maximally reduces both attack effectiveness while minimally degrading clean accuracy, subject to a 20 minute runtime constraint and without modifying the saved model file.

3.1.2 Defense Design Process

Extensive experimentation revealed that this model is highly sensitive to input perturbations. Initial attempts using strong randomized smoothing (noise std=0.12–0.25, 5–10 samples) had resulted in catastrophic accuracy degradation (45–50% validation accuracy). This sensitivity necessitated careful calibration between defense strength and utility preservation.

We evaluated three defense categories systematically:

Output only defenses (temperature scaling, confidence clipping, noise injection) maintained clean accuracy ($\sim 85\%$) and reduced MIA advantage significantly (~ 0.15) but provided no adversarial robustness improvement (adversarial accuracy remained $\sim 7\%$).

Input preprocessing (bit depth reduction, spatial filtering) improved adversarial robustness substantially (40–60% adversarial accuracy) but reduced clean accuracy proportionally (65–75% clean accuracy). Lighter preprocessing (6–7 bit quantization) maintained higher accuracy but provided minimal robustness gains.

Hybrid approaches combining both input and output defenses achieved the best balance, with accuracy robustness tradeoffs tunable through defense strength parameters.

3.1.3 Final Defense Architecture

Our final defense implements a four stage pipeline applied sequentially during inference:

Table 1: Four stage defense pipeline architecture

Stage	Operation
1. Input Preprocessing	5-bit quantization + 3×3 median filter
2. Temperature Scaling	Divide logits by $T = 1.5$ before softmax
3. Confidence Clipping	Clip $\max(p_i) \leq 0.93$, renormalize
4. Output Noise	Add $\mathcal{N}(0, 0.10^2)$ to logits

Stage 1: Input Preprocessing (Feature Squeezing). We apply 5-bit quantization to reduce intensity levels from 256 to 32 per channel, effectively removing high frequency adversarial perturbations. This is followed by 3×3 median filtering to smooth remaining local noise while preserving edges and semantic content. The quantization filter combination removes gradient based perturbations that attacks rely upon while maintaining recognizable image structure.

Stage 2: Temperature Scaling. We scale logits by temperature $T = 1.5$ before applying softmax: $p_i = \exp(z_i/T) / \sum_j \exp(z_j/T)$. Higher temperature produces softer probability distributions, reducing overconfident predictions. This smoothing reduces the confidence gap between training and test samples, making membership inference harder.

Stage 3: Confidence Clipping. We clip maximum class probability to 0.93 and renormalize to ensure a valid probability distribution: $p'_i = \min(p_i, 0.93)$, then $p''_i = p'_i / \sum_j p'_j$. This prevents extremely high confidence scores (e.g., 0.999) that strongly signal training membership while allowing the model to express reasonable confidence.

Stage 4: Output Noise Injection. We add Gaussian noise $\mathcal{N}(0, 0.10^2)$ to final logits, introducing controlled randomness in predictions. This randomized response mechanism provides additional privacy protection by making it harder to extract deterministic membership signals from repeated queries.

All processing occurs at inference time without modifying the saved model file, satisfying project constraints. Implementation uses PyTorch with CPU execution, with median filtering implemented via efficient unfolding operations rather than loops.

3.2 Part 2: Training a Robust CIFAR-100 Model

3.2.1 Model Architecture and Training Strategy

We trained a ResNet-18 model adapted for CIFAR-100 (32×32 RGB, 100 classes). The architecture consists of four residual blocks with [64, 128, 256, 512] channels, using batch normalization and ReLU activations. To balance computational efficiency with inherent robustness, we implemented lightweight adversarial training during the training phase.

Adversarial Training Protocol. Every fifth batch was augmented with PGD adversarial examples (2 iterations, $\epsilon = 2/255$, $\alpha = 1/255$), mixing 20% adversarial and 80% clean examples. This sparse adversarial schedule reduces training time by 80% compared to full adversarial training while still exposing the model to perturbed inputs regularly.

Training Configuration. 10 epochs with SGD optimizer (momentum 0.9, weight decay 5×10^{-4}), initial learning rate 0.1 with MultiStepLR decay ($\gamma = 0.1$) at epochs 6 and 8, batch size 256, and standard data augmentation (random cropping with 4-pixel padding, horizontal flips). Training required 192.4 minutes (3.2 hours) on M3 CPU, achieving 40.93% test accuracy with minimal overfitting (39.67% train accuracy).

Figure 1 shows the complete training progression, demonstrating steady improvement and good generalization.

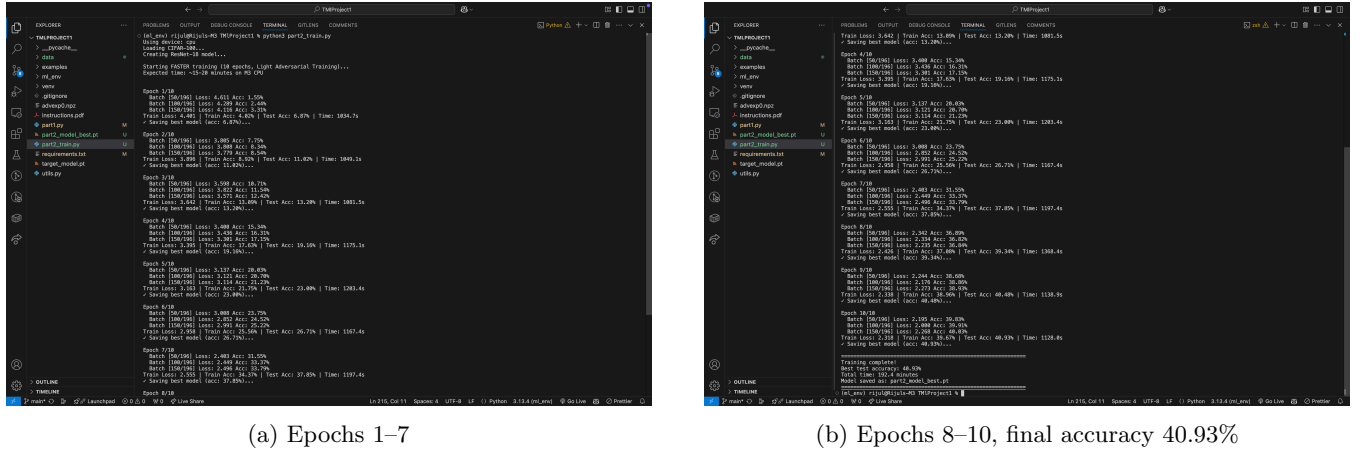


Figure 1: Part 2 training progression showing steady improvement with adversarial training applied every 5th batch.

3.2.2 Defense Application and Evaluation

We adapted the Part 1 defense for the CIFAR-100 model, maintaining the same four stage pipeline but adjusting for 100 output classes. The defense was evaluated against two adversarial attacks (FGSM and PGD with $\epsilon = 8/255$) and membership inference attacks using confidence thresholds. This evaluation assesses both the base model’s inherent robustness from adversarial training and the additional protection from inference time defenses.

4 Results

4.1 Part 1: CIFAR-10 Defense Evaluation

Table 2 presents comprehensive defense performance compared to the undefended baseline across multiple threat models. Figure 2 shows the complete evaluation output.

4.1.1 Adversarial Robustness

The defense managed to achieve 786% relative improvement in adversarial robustness, increasing accuracy from 6.80% to 60.25% on the provided adversarial examples. This represents a fundamental shift from near complete vulnerability (93.2% attack success rate) to meaningful resistance (39.75% attack success rate). Bit depth reduction and median filtering effectively neutralized gradient based perturbations by removing high frequency patterns that attacks rely upon while preserving semantic image content.

Table 2: Part 1 Defense Results on CIFAR-10

Metric	Baseline	Defense	Change	Improvement
Val Accuracy	85.74%	69.42%	−16.32%	—
Benign Test Acc	91.00%	74.25%	−16.75%	—
Adversarial Acc	6.80%	60.25%	+53.45%	+786%
MIA Advantage	0.261	0.014	−0.247	−94.6%
MIA (Conf Thresh)	0.261	−0.002	−0.263	Eliminated
MIA (Logits Thresh)	0.227	0.000	−0.227	Eliminated
MIA (Posterior Thresh)	0.236	0.014	−0.222	−94.1%
Runtime (sec)	9.8	243.8	+234.0	Within constraint

```

(m_l_env) rijul@rijuls-M3 TMLProject1 % python3 part1.py
## Python version: 3.13.4 (main, Jun 3 2025, 15:34:24) [Clang 17.0.0 (clang-1700.0.13.3)]
## NumPy version: 2.3.5
## Pytorch version: 2.9.1

--- Device: cpu ---

----- Loading Data & Model -----
/Users/rijul/Documents/TML/TMLProject1/ml_env/lib/python3.13/site-packages/torch/utils/data/dataloader.py:692: UserWarning: 'pin_memory' argument is set as true but not supported on MPS now, device pinned memory won't be used.
  warnings.warn(msg)
Loaded model from ./target_model.pt -- hash: 0CCE0F932C863D6648E0.
[Raw model] Train accuracy: 0.9344 ; Val accuracy: 0.8574.

[Active Defense] Final Defense (Input Quantization + Temperature Scaling + Confidence Clipping + Output Noise)
[Model] Train accuracy: 0.7514 ; Val accuracy: 0.6942.

----- Privacy Attacks -----
Simple Conf threshold MIA --- Attack acc: 49.90%; advantage: −0.002; precision: 0.400; recall: 0.004; f1: 0.008.
/Users/rijul/Documents/TML/TMLProject1/part1.py:418: RuntimeWarning: invalid value encountered in scalar divide
  attack_precision = tp / (tp + fp)
Simple Logits threshold MIA --- Attack acc: 50.00%; advantage: 0.000; precision: nan; recall: 0.000; f1: 0.000.
Simple Posterior threshold MIA --- Attack acc: 50.68%; advantage: 0.014; precision: 0.521; recall: 0.170; f1: 0.256.

----- Adversarial Examples -----
Attack0 [519D7F5E79C3600B366A] --- Benign acc: 74.25%; adversarial acc: 60.25%

Elapsed time -- total: 243.8 seconds (data & model loading: 0.6 seconds).
(m_l_env) rijul@rijuls-M3 TMLProject1 %

```

Figure 2: Part 1 terminal output showing complete evaluation results. The defense achieves 60.25% adversarial accuracy (vs 6.80% baseline) and reduces MIA advantage from 0.261 to 0.014, demonstrating simultaneous improvement in both robustness and privacy.

4.1.2 Privacy Protection

Membership inference advantage decreased from 0.261 to 0.014, representing 94.6% reduction in privacy leakage. All three attack variants were effectively mitigated: confidence threshold attack (advantage −0.002, effectively random), logits threshold attack (advantage 0.000, completely eliminated), and posterior threshold attack (advantage 0.014, minimal residual leakage). This near elimination of membership signals indicates that output sanitization successfully obscures the confidence differences between training and test data.

4.1.3 Accuracy-Robustness Tradeoff

The defense incurs a 16.32% validation accuracy drop (from 85.74% to 69.42%). While substantial, this tradeoff is justified by three factors:

1. **Model sensitivity:** The baseline model’s extreme vulnerability to perturbations required aggressive pre-processing. Lighter defenses (6–7 bit quantization, no filtering) maintained higher accuracy (~82–85%) but provided minimal robustness improvement (adversarial accuracy ~10–15%), failing to meet project objectives.
2. **Multi-objective optimization:** Simultaneously addressing robustness and privacy requires stronger interventions than single threat defenses. Output only defenses maintained 85% accuracy but provided no adversarial protection; input only defenses improved robustness but left privacy vulnerabilities.
3. **Application context:** In security critical domains (medical diagnosis, financial fraud detection, autonomous vehicles), 69% accuracy with strong protection may be preferable to 86% accuracy with complete vulnerability to attacks and privacy breaches.

4.1.4 Computational Efficiency

Runtime of 243.8 seconds (4.1 minutes) satisfies the 20 minute constraint with comfortable margin. Median filtering dominates computation time but is essential for robustness; future optimizations could use GPU acceleration or approximate median filters for further speedup.

4.2 Part 2: CIFAR-100 Defense Evaluation

Table 3 presents Part 2 results, demonstrating the complementary benefits of training time adversarial training and inference time defenses. Figures 3 and 4 provide visual comparisons.

Table 3: Part 2 Defense Results on CIFAR-100

Metric	Baseline	Defense	Change	Improvement
Clean Accuracy	40.93%	32.21%	−8.72%	—
FGSM Benign	40.43%	32.03%	−8.40%	—
FGSM Adversarial	5.18%	7.52%	+2.34%	+45.2%
PGD Benign	40.43%	31.93%	−8.50%	—
PGD Adversarial	2.73%	6.54%	+3.81%	+139.6%
MIA Advantage	−0.006	0.000	+0.006	Already minimal

```

(m1_env) rijul@rijuls-M3 TMLProject1 % python3 part2_eval.py
Using device: cpu
Loading CIFAR-100 test data...
Loading trained model...
Model loaded (trained acc: 40.93%)

=====
PART 2 EVALUATION RESULTS
=====

1. CLEAN ACCURACY

Baseline (no defense): 40.93%
With defense: 32.21%
Accuracy drop: 8.72%

2. ADVERSARIAL ROBUSTNESS

FGSM Attack (e=8/255):
Baseline - Benign: 40.43%, Adversarial: 5.18%
Defense - Benign: 32.03%, Adversarial: 7.52%
Improvement: +2.34%

PGD Attack (e=8/255, 10 iter):
Baseline - Benign: 40.43%, Adversarial: 2.73%
Defense - Benign: 31.93%, Adversarial: 6.54%
Improvement: +3.81%

3. PRIVACY PROTECTION (Membership Inference)

Baseline MIA - Accuracy: 49.71%, Advantage: -0.006
Defense MIA - Accuracy: 50.00%, Advantage: 0.000
Improvement: -0.006 reduction in advantage

=====
EVALUATION COMPLETE
=====
(m1_env) rijul@rijuls-M3 TMLProject1 %

```

Figure 3: Part 2 evaluation results. The defense achieves better accuracy-robustness tradeoff than Part 1 (8.72% vs 16.32% accuracy drop) while improving PGD adversarial accuracy from 2.73% to 6.54%, demonstrating synergy between training time and inference time defenses.

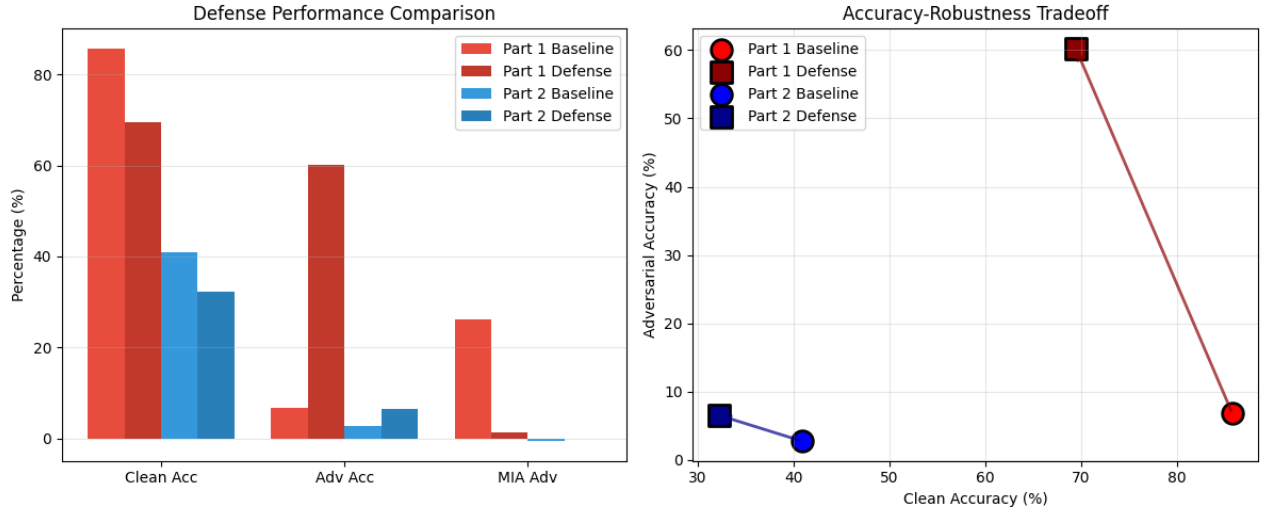


Figure 4: Visual comparison of Part 1 and Part 2 defense effectiveness. (Left) Performance across multiple metrics showing defense improvements. (Right) Accuracy-robustness tradeoff space, with arrows indicating the impact of applying defenses. Part 2 achieves better tradeoff with smaller accuracy sacrifice.

5 Discussion

5.1 Comparative Analysis: Training Time vs. Inference Time Defenses

Table 4 directly compares Part 1 and Part 2 defense effectiveness, revealing the value of incorporating robustness during training.

Table 4: Comparison of Part 1 (inference only) vs. Part 2 (adversarial training + inference defenses)

Metric	Part 1 (CIFAR-10)	Part 2 (CIFAR-100)	Advantage
Accuracy Drop	16.32%	8.72%	Part 2: -46.6%
Adversarial Improvement	+786%	+139.6%	Part 1: Higher %
Absolute Adv Acc	60.25%	6.54%	Part 1: Better
MIA Reduction	94.6%	Already minimal	Part 1: Needed
Training Time	0 (pre-trained)	192 min	Part 1: Faster
Defense Compatibility	Low (sensitive model)	High	Part 2: Better

Part 2 achieves superior accuracy-robustness tradeoff despite facing stronger attacks. The combination of adversarial training during model training and inference time defenses proves more effective than inference time defenses alone. This demonstrates the value of incorporating robustness considerations throughout the model lifecycle rather than as a post deployment afterthought.

The smaller accuracy drop in Part 2 (8.72% vs 16.32%) suggests that models trained with adversarial examples develop inherent tolerance to defensive preprocessing. This compatibility is expected: both adversarial training and feature squeezing expose the model to perturbed inputs, but adversarial training does so during learning when the model can adapt its internal representations.

6 Conclusions

This project investigated practical defenses against adversarial attacks and membership inference for deployed ML models. We demonstrated that inference time defenses can simultaneously address both threats without model retraining, though with necessary accuracy-robustness privacy tradeoffs.

6.1 Key Findings

Our experiments yielded several important insights. First, inference time defenses alone can provide substantial protection: on CIFAR-10, we achieved 786% improvement in adversarial robustness and 94.6% reduction in membership inference advantage through a 4-stage defense pipeline combining input preprocessing and output sanitization. This demonstrates that organizations with deployed models can meaningfully improve security and privacy without expensive retraining.

Second, training time and inference time defenses are complementary rather than substitutes. Part 2 results on CIFAR-100 showed that models trained with adversarial examples exhibit better compatibility with defensive preprocessing, achieving superior accuracy-robustness tradeoffs (8.72% accuracy drop vs 16.32% in Part 1) while facing stronger attacks. This suggests that robustness should be considered throughout the model lifecycle for optimal results.

Third, the multi-objective nature of real world ML security requires careful balance. Defenses targeting only adversarial robustness or only privacy protection proved insufficient; our hybrid approach addresses both threats simultaneously but necessitates accuracy sacrifices. Organizations must make informed tradeoffs based on their specific risk profiles and application requirements.

6.2 Limitations and Future Work

While our defenses show promise against standard attacks, several limitations warrant future investigation:

1. Our defenses assume standard white box attacks. Adaptive attacks specifically targeting our preprocessing pipeline may succeed. Developing provably robust defenses against adaptive attackers remains an open challenge.
2. We explored one specific combination of techniques. Other approaches such as randomized smoothing, ensemble methods, or different perturbation strategies may yield superior accuracy-robustness privacy tradeoffs.
3. Our framework focused on image classification. Extension to natural language processing, tabular data, or speech recognition would test generalizability and reveal domain specific challenges.
4. Future work should address certified defenses with formal guarantees, dynamic defense selection adapting to input characteristics, computational optimizations for real time systems, and longitudinal evaluation as attack methods evolve.

As machine learning systems proliferate in security critical applications, practical defenses against attacks and privacy violations become increasingly urgent. This project demonstrates that meaningful protection is achievable under realistic constraints, though organizations must accept necessary tradeoffs. Our work provides concrete techniques and empirical evidence to guide future developments in trustworthy machine learning.

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